Random ticks can cause entity build-up in lazy chunks

MC-280155 `Chunk loading`

Status	Resolved / Fixed
Affected	25w06a
Fixed in	25w09a
Reported	2025-02-06

STEPS TO REPRODUCE

1. Create an empty or superflat Creative world.
2. Run the following commands: `/gamerule doMobSpawning false`, `/gamerule spawnChunkRadius 0`, and `/tp @e[type=!minecraft:player] ~ -1000 ~`.
3. Set up the entity-count scoreboard by running `/scoreboard objectives add entities dummy` and `/scoreboard objectives setdisplay sidebar entities`.
4. Enable chunk borders with F3 + G, stand in the middle of a chunk, place a block to mark it, and run `/forceload add ~ ~`.
5. Move to an adjacent chunk and place sand with a cactus on top; place a stable block diagonally one block upward and one block adjacent to the cactus in a cardinal direction.
6. Enter spectator mode with F3 + N, then run `/gamerule spectatorsGenerateChunks false`.
7. Increase the random tick speed by running `/gamerule randomTickSpeed 3000`.
8. Wait until the cactus randomly grows and breaks, causing cactus item entities to appear.
9. Freeze the game, set the random tick speed to zero, record the item count, advance 6000 ticks, and record the count again by running `/tick freeze`, `/gamerule randomTickSpeed 0`, `/execute store result score @s entities if entity @e[type=item, nbt={"Item":{"id":"minecraft:cactus"}}]`, `/tick sprint 6000`, and `/execute store result score @s entities if entity @e[type=item, nbt={"Item":{"id":"minecraft:cactus"}}]`.
10. Observe that the cactus items remain present in the same quantity after 6000 ticks, indicating that they have not aged or despawned while in the non-entity-processing chunk.

OBSERVED BEHAVIOUR

Random ticks are processed in the adjacent loaded chunk even though it is outside the entity-processing range. As a result, the cactus continues to grow while the chunk is in a lazy, non-entity-processing state.

When the cactus grows into the adjacent blocking block, it breaks and creates a cactus item entity. Repeated random ticks cause additional cactus item entities to be created in the same chunk. These item entities remain present while the chunk is not entity-processing: their item age does not advance, so they do not reach the normal 6000-tick despawn threshold.

After advancing the game by 6000 ticks, the number of cactus item entities remains unchanged. They only begin aging and become eligible for despawning once the chunk enters entity-processing range, such as when a player moves close enough to process entities in that chunk. This allows item entities to accumulate indefinitely in loaded lazy chunks and can produce excessive entity counts and associated server performance problems.

EXPECTED BEHAVIOUR

Cacti and other random-tick-dependent blocks in lazy, non-entity-processing chunks should not update in a way that creates item entities. In this test, the cactus should remain intact, no cactus items should accumulate, and the entity count should not increase while the chunk remains lazy. Item entities should also age and despawn normally once processed.

ENVIRONMENT

Minecraft Java Edition Version 25w06a (snapshot)

ORIGINAL DESCRIPTION

Summary:

Due to the new changes with random ticks in loaded chunks away from the player, item entities can be created due to a random tick interaction in a non-entity-processing chunk and will not begin to despawn until the chunk they are in becomes entity-processing. This could lead to undesirable and accidental item build-up that may cause high MSPT or crash servers. A realistically common example of this may be a chunk being loaded for an extended duration while being unintentionally near a player-built cactus, sugar cane, pumpkin/melon, or bamboo farm that automatically harvests the crops through any block-based interactions (such as pistons extending, or cacti naturally popping off when grown adjacent to a block).

This seems unintentional and could be especially dangerous in the edge case where a cactus may have naturally generated within the lazy spawn chunks in a way that causes it to break upon growth due to the surrounding terrain or structures.

Suggested Fix:

Allowing random ticks to only occur within entity-processing chunks would greatly reduce the chances of this happening and better align with the spirit of the change.

Steps To Reproduce:

1. Create an empty or superflat creative world and use all of the following commands to ensure a clean test:

/gamerule doMobSpawning false

/gamerule spawnChunkRadius 0

/tp @e[type=!minecraft:player] ~ -1000 ~

2. Set up a dummy scoreboard objective to track the number of loaded entities as follows:

/scoreboard objectives add entities dummy

/scoreboard objectives setdisplay sidebar entities

3. Enable the Chunk borders display using F3 + G and Stand somewhere within the middle of a chunk. Place an easily discernible block to mark this chunk so you don't lose it.

Forceload the chunk you are standing in using the command:

/forceload add ~ ~

4. Move to the next chunk over in any cardinal or ordinal direction and place a block of sand with a cactus planted on top and a stable block placed diagonally one block upwards and one block adjacent in any cardinal direction from the cactus (an image of this setup is attached below)

5. Enter spectator mode via F3 + N and use the command:

/gamerule spectatorsGenerat

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: https://bugs.mojang.com/browse/MC/issues/MC-280155